

DTSA 5744

Symmetric Cryptography

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (50.2 / 100)	39.1 / 100
SPECIALIZATION Applied Cryptography		

39

CURRENT AVG SCORE
across 6 job roles

81

POTENTIAL SCORE
with recommended additions

+53

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course teaches the principles, requirements, constructions, and applications of symmetric cryptography based on the shared secret key. The course describes substitution and transposition techniques which have been invented before computers. Then, we build on the product ciphers to learn about the modern block ciphers and review the popular cipher algorithms of DES, 3-DES, and AES. We also describe the block cipher operation modes to enable the block cipher to support variable data length.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	50.2	88.5	+38.3	HIGH
Applied Scientist	39.5	79.8	+40.4	HIGH
ML Engineer	36.9	68.4	+31.5	HIGH
AI Engineer	36.3	80.6	+44.3	HIGH
MLOps Engineer	36.1	88.8	+52.7	HIGH
Data Scientist	35.5	77.8	+42.4	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

50.2 → 88.5 (+38.3 pts)

sets learning tools protection business design science models

Applied Scientist

39.5 → 79.8 (+40.4 pts)

computational problems models technical learning systems design studies

ML Engineer

36.9 → 68.4 (+31.5 pts)

learning machine systems models model infrastructure problems impact

AI Engineer

36.3 → 80.6 (+44.3 pts)

models learning infrastructure software technical code model systems

MLOps Engineer

36.1 → 88.8 (+52.7 pts)

infrastructure detection model text systems learning machine image

Data Scientist

35.5 → 77.8 (+42.4 pts)

sets presentation mitigation different problems approaches

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.